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Application of social force model to pedestrian behavior analysis at signalized crosswalk

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Highlights

- Formulate pedestrians' decision-making behavior at signalized crosswalk.
- Represent pedestrian evasion behavior and collision avoidance with vehicles.
- Calibrate model parameters by separating into measurable and non-measurable groups.
- Visually reproduce pedestrian crossing behavior in microscopic simulation.

Abstract

Limited pedestrian behavior models shed light on the case at signalized crosswalk, where pedestrian behavior is characterized by group or individual evasion with surrounding pedestrians, collision avoidance with conflicting vehicles, and response to signal control and crosswalk boundary. This study fills this gap by developing a microscopic simulation model for pedestrian behavior analysis at signalized intersection. The social force theory has been employed and adjusted for this purpose. The parameters, including measurable and non-measurable ones, are either directly estimated based on observed dataset or indirectly derived by maximum likelihood estimation. Last, the model performance was confirmed in light of individual trajectory comparison between estimation and observation, passing position distribution at several cross-sections, collision avoidance behavior with conflicting vehicles, and lane-formation phenomenon. The simulation results also concluded that the model enables to visually represent pedestrian crossing behavior as in the real world.

Previous Next

Keywords

Pedestrian behavior; Social force; Microscopic traffic simulation; Signalized intersection

1. Introduction

The importance of modeling pedestrian behavior has been well recognized in both fields of safety and capacity assessment of road facilities (Papadimitriou et al., 2009). In this regard, simulation can be a powerful tool for pedestrian behavior reproduction. It has been successfully applied to human navigation analysis (Usher and Strawderman, 2010, Kneidl et al., 2013), pedestrian facility assessment (Zhang et al., 2008, Maria et al., 2013) and crowd dynamic analysis (Hoogendoorn and Daamen, 2005, Helbing et al., 2005). The simulation technology offers a flexible approach for detailed analysis of pedestrian behavior that may result in safety or efficiency problem at road facilities.

This study is part of intensive efforts to develop a microscopic simulation model for safety assessment at signalized intersection. Although signalized crosswalks are operated to give pedestrians prioritized right of way, more than 30% of the total traffic accident fatalities in Japan are pedestrians at crosswalks (National Police Agency of Japan, 2013) whereas 15% in Germany (German Institute for Economic Research, 2010). Many reasons exist behind such statistics, such as risk-taking behavior, intersection geometry, and traffic signal control. Among these factors, stochastic pedestrian behavior is deemed as a critical factor that may result in safety problems. In order to develop such a simulation model, pedestrian behavior must be reasonably represented.

To date, pedestrian behavior modeling has attracted considerable attention. With comparison to macroscopic models, microscopic models provide a more detailed description of pedestrian behavior. Generally, existing microscopic models can be classified into two categories: discrete and continuum models.

The family of discrete models essentially includes discrete choice model (Antonini et al., 2006, Robin et al., 2009, Guo and Huang, 2012), lattice gas model (Helbing et al., 2003, Guo and Huang, 2008) and cellular automaton model (Burstedde et al., 2001, Kirchner and Schadschneider, 2002), in which space is discretized to approximate real pedestrian movement. It is convenient to set various rules in a discrete model, such as direction keeping, destination orientation, free-flow acceleration, leader–follower maneuver, and collision avoidance. However, only limited choices for speed and direction in a discrete choice model and limited cellular alternatives in a cellular automaton model are provided, which is difficult to represent the flexible pedestrian behavior, especially for the interactive case in the real world.

Continuum models use differential equations to describe the dynamic movement in space. At the early stage, a magnetic-force model was developed by borrowing a motion equation used for magnetic fields (Okazaki, 1979). Based on this concept, a more robust physical force based model, i.e., social force model, was applied to evacuation analysis (Helbing and Molnar, 1995). A comparison overview of crowd motion simulation models shows that the force based model has many advantages (Duives et al., 2013). Firstly, it enables to describe the dynamic movement of pedestrian in space and time. Secondly, each equation and parameter can be easily explained from the viewpoint of kinematics. Thirdly, it allows reproducing the typical features of pedestrian flow, such as lane formation and oscillation, due to the interaction with other pedestrians (Helbing et al., 2000, Helbing et al., 2001). Although numerous studies based on social force model focused on the general mechanism of pedestrian behavior, few of them shed light on the application of social force model to pedestrian behavior at signalized crosswalk. Teknomo (2006) developed a social force based model and analyzed the pedestrian behavior of one-way and two-way pedestrian flow at crosswalk. It was found that two-way traffic performance reduces significantly as the number of pedestrians increase. Li et al. (2012) modified the social force model by considering the required space and the critical gaps with turning vehicles, which makes it possible to describe the stop/go decision to the conflicting vehicles. However, these models, neither fully developed nor calibrated based on real trajectory data, failed to illustrate the crucial pedestrian behavior at crosswalk.

To fill the gap, the special characteristics of pedestrian behavior at signalized crosswalk are considered in this study. It takes into account the crossing behaviors such as group evasive and individual evasive behavior with surrounding pedestrians, collision avoidance with turning vehicles, reaction to crosswalk boundary, and response to signal control, which are all important to

evaluate the safety performance of signalized intersections. Then, a statistical method is applied to parameter calibration. In this study, the parameters are separated into two groups, measurable and non-measurable ones. The measurable parameters are directly estimated from the observed trajectory dataset, and the non-measurable parameters are derived by using maximum likelihood estimation (MLE). Last, the results of simulation scenarios conclude that the model enables to represent the typical characteristics of pedestrian behavior at signalized crosswalk such as lane-formation phenomenon, collision avoidance with conflicting vehicles and surrounding pedestrians.

2. Model framework

As shown in Fig. 1, the general structure of the model is based on the decision making level of psychological process. The first level is strategic level, at which pedestrians make a decision of the intended direction. The second level is operational level, which is the second step for pedestrians to adjust their walking behavior at the crosswalk. Note that after reacting to the pedestrian signal phase and deciding to enter the crosswalk, pedestrians will adjust their speeds and directions dynamically when interacting with other road users and crosswalk facilities.

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Fig. 1. Psychological process of pedestrian behavior.

3. Desired direction model

The psychological process of decision-making of desired direction is like the route-choice strategy. As shown in Fig. 2, there are four OD zones (N1, N2, F1, F2) defined for pedestrian crossing. Accordingly, there are three stages for a pedestrian finishing the crossing, i.e., from origin to crosswalk entering position, from crosswalk entering position to crosswalk exit position, and last from crosswalk exit position to destination. It is assumed that the desired entering position at crosswalk is determined once the pedestrians enter the waiting area and the desired exit position is determined once the pedestrian enter the crosswalk. Thus, the desired direction at waiting area (stage ①) is defined as the vector pointing from current position to the desired entering position, and the desired direction at crosswalk (stage ②) is defined as the vector pointing from the current position to the desired exit position. Since the focus of this study is pedestrian walking behavior at crosswalk, only the second stage is discussed in the following sections.

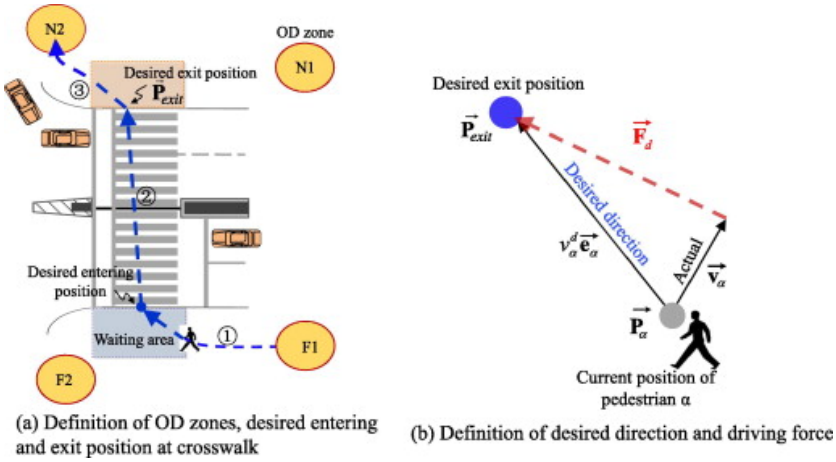

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Fig. 2. Definition of OD and desired direction.

Based on the empirical analysis of pedestrian trajectory, the entering and exit positions at the edge of crosswalk were found to be influenced by crosswalk geometry, pedestrian origin–destination movement, previous passing position and the densities of other road users. Considering that Weibull distribution is able to represent various types of distribution including skewed ones, it is applied to estimate the desired entering and exit positions of pedestrians. Thus, the probability density function of passing position distribution is established by using Weibull distribution as shown in Eq. (1) (Zeng et al., 2013).

$$f(x;\alpha,\beta) = \frac{\alpha}{\beta}\left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^{\alpha}} \tag{1}$$
$$\alpha = f(y_{1,1},y_{1,2},\ldots,y_{1,n}) = \lambda_{1,1}y_{1,1} + \lambda_{1,2}y_{1,2} + \cdots + \lambda_{1,n}y_{1,n} + \lambda_{1,n+1}$$
$$\beta = f(y_{2,1},y_{2,2},\ldots,y_{2,n}) = \lambda_{2,1}y_{2,1} + \lambda_{2,2}y_{2,2} + \cdots + \lambda_{2,n}y_{2,n} + \lambda_{2,n+1}$$

where $f(x;\alpha,\beta)$ is the probability density of passing position distribution; x is the position at entering or exit cross-section; $y_{1,1},\ldots,y_{2,n}$ are independent variables of influencing factors such as crosswalk width, origin–destination movement, and road user density; $\lambda_{1,1},\ldots,\lambda_{2,n+1}$ are model coefficient estimated by the maximum likelihood method.

4. Stop/go decision model

It is assumed that all the pedestrians choose to go when the signal phase turns to green and to stop entering the crosswalk once the signal phase turns to red. However, pedestrian flashing green (PFG) is a special signal phase which is considered as a clearance time interval for pedestrians to safely finish walking at crosswalk. In Japan, although pedestrians are not supposed to enter the crosswalk during PFG by law, many of them rush into the crosswalk without taking care of approaching turning vehicles, which may cause higher probability to get severe conflicts with the vehicles. Thus, a stop/go decision model is necessary to present the pedestrian crossing behavior at PFG. According to the pedestrian crossing behavior model by Asano et al. (2013), the probability of stop/go is related to the walking speed and the distance from the current position to the crosswalk at the onset of PFG. In the binary Logit-based model, the error term of utility function to choose “go” is assumed to follow Gumbel distribution and the probability to go is represented by Eq. (2).

$$P_r(go) = \frac{\exp(V)}{1+\exp(V)} \tag{2}$$
$$V = f(y_{1,1},y_{1,2},\ldots,y_{1,n}) = \lambda_{1,1}y_{1,1} + \lambda_{1,2}y_{1,2} + \cdots + \lambda_{1,n}y_{1,n} + \lambda_{1,n+1}$$

where $y_{1,1},\ldots,y_{1,n}$ is the independent variables of influencing factors such as the distance to crosswalk, the instant speed at the onset of PFG and the movement direction; $\lambda_{1,1},\ldots,\lambda_{1,n+1}$ are model coefficients estimated by the maximum likelihood method.

5. The modified social force model

The modified social force model in this study relates five sources of social force: driving force toward destination $(\vec{\mathbf{F}}_D)$, repulsive or attractive force from crosswalk boundary $(\vec{\mathbf{F}}_B)$, repulsive force from surrounding pedestrians $(\vec{\mathbf{F}}_P)$, repulsive force from conflicting vehicles $(\vec{\mathbf{F}}_V)$, and attractive force from signal phase $(\vec{\mathbf{F}}_S)$. The resultant force $(\vec{\mathbf{F}}(t_k))$ can be expressed by Eq. (3).

$$\vec{\mathbf{F}}(t_k) = \vec{\mathbf{F}}_D + \vec{\mathbf{F}}_B + \vec{\mathbf{F}}_P + \vec{\mathbf{F}}_V + \vec{\mathbf{F}}_V \tag{3}$$

The resultant speed vector $\left(\vec{\mathbf{v}}_r(t_k)\right)$ can be expressed by Eq. (4).

$$\vec{\mathbf{v}}_r(t_k) = \vec{\mathbf{v}}_\alpha(t_k) + \vec{\mathbf{F}}(t_k)\Delta t \tag{4}$$

where $\vec{\mathbf{v}}_\alpha(t_k)$ is the current speed vector at time t_k, Δt is the time step in simulation. Finally, the updated position in simulation is given by Eq. (5).

$$\vec{\mathbf{P}}_\alpha(t_{k+1}) = \vec{\mathbf{P}}_\alpha(t_k) + \frac{\vec{\mathbf{v}}_\alpha(t_k)+\vec{\mathbf{v}}_r(t_k)}{2}\Delta t \tag{5}$$

where $\vec{\mathbf{P}}_\alpha(t_k)$ is the current position at time t_k, and $\vec{\mathbf{P}}_\alpha(t_{k+1})$ is the updated position of the next time step t_{k+1}.

5.1. Driving force

The desired direction is defined by the current position and the exit position at crosswalk, toward which pedestrians desire to walk. However, pedestrians change their speed vector dynamically due to the stimulus of surrounding environment. As shown in Fig. 2, a deviation of the current speed vector from the desired speed vector leads to a force to recover to the desired speed vector again within a certain relaxation time. This force is called driving force as shown in Eq. (6).

$$\vec{\mathbf{F}}_D = \frac{1}{\tau_\alpha}\left(\nu_\alpha^d \vec{\mathbf{e}}_\alpha - \vec{\mathbf{v}}_\alpha\right) \tag{6}$$

where ν_α^d is the quantity of the desired speed, $\vec{\mathbf{e}}_\alpha$ is the unit vector of the desired direction, and τ_α is the acceleration time of pedestrian α. According to the empirical analysis of pedestrian speed at signalized crosswalk, the desired speed ν_α^d is assumed to follow Gamma distribution, which can be estimated by entering speed, crosswalk length and the elapsed time of PG. The cumulative Gamma distribution function for desired speed is shown as Eq. (7) (Zhang et al., 2013).

$$\begin{aligned} &P(\nu_\alpha^d = x) = \frac{1}{\beta^\alpha \Gamma(\alpha)}\exp\left(-\frac{x-\gamma}{\beta}\right)(x-\gamma)^{\alpha-1} \\ &\alpha = f(y_{1,1}, y_{1,2}, \ldots, y_{1,n}) = \lambda_{1,1}y_{1,1} + \lambda_{1,2}y_{1,2} + \cdots + \lambda_{1,n}y_{1,n} \\ &+ \lambda_{1,n+1} \\ &\beta = f(y_{2,1}, y_{2,2}, \ldots, y_{2,n}) = \lambda_{2,1}y_{2,1} + \lambda_{2,2}y_{2,2} + \cdots + \lambda_{2,n}y_{2,n} \\ &+ \lambda_{2,n+1} \\ &\gamma = f(y_{3,1}, y_{3,2}, \ldots, y_{3,n}) = \lambda_{3,1}y_{3,1} + \lambda_{3,2}y_{3,2} + \cdots + \lambda_{3,n}y_{3,n} \\ &+ \lambda_{3,n+1} \end{aligned} \tag{7}$$

where $\Gamma(\alpha)$ is Gamma function; x is the random desired speed; $y_{1,1},\ldots,y_{3,n}$ are independent variables of influencing factors such as entering speed, crosswalk length and the elapsed time of PG; $\lambda_{1,1},\ldots,\lambda_{3,n+1}$ are model coefficient estimated by the maximum likelihood method.

5.2. Force from crosswalk boundary

Generally, pedestrians keep walking inside the boundary of crosswalk. As shown in Fig. 3(a), it is assumed that when there is a velocity component $\left(\vec{\mathbf{v}}_\alpha^x\right)$ toward the crosswalk boundary, a repulsive force $\left(\vec{\mathbf{F}}_{B\alpha}^r\right)$ will be generated. The repulsive force makes the pedestrian keep a certain distance from the boundary. However, when the pedestrian density increases to some extent, some pedestrians may walk outside the crosswalk to avoid serious conflicts with others. However, most of them will

move back to the crosswalk once the conflict becomes less severe. Thus, as shown in Fig. 3(b), it is assumed that an attractive force $\left(\vec{\mathbf{F}}^{a}_{B\alpha}\right)$ will attract back those pedestrians outside the crosswalk. The social force from crosswalk boundary $\left(\vec{\mathbf{F}}_{B}\right)$ can be presented by Eq. (8).

$$\vec{\mathbf{F}}_{B} = \begin{cases} \vec{\mathbf{F}}^{r}_{B\alpha} = A^{r}_{B}\exp\left(-B^{r}_{B}\left\|\vec{\mathbf{P}}_{\alpha} - \vec{\mathbf{P}}_{B}\right\|\right)\vec{\mathbf{n}}_{B\alpha}, & \text{if pedestrian } \alpha \text{ is inside the crosswalk} \\ \vec{\mathbf{F}}^{a}_{B\alpha} = A^{a}_{B}\exp\left(B^{a}_{B}\left\|\vec{\mathbf{P}}_{\alpha} - \vec{\mathbf{P}}_{B}\right\|\right)\vec{\mathbf{n}}_{\alpha B}, & \text{otherwise} \end{cases} \tag{8}$$

where $\left(\vec{\mathbf{P}}_{B}\right)$ is the perpendicular foot point of pedestrian α on the nearest crosswalk boundary, $\left(\vec{\mathbf{n}}_{B\alpha}\right)$ is the normalized vector pointing from boundary to pedestrian α, $\left(\vec{\mathbf{n}}_{B\alpha}\right)$ is the normalized vector pointing from pedestrian α to crosswalk boundary. A^{r}_{B}, A^{a}_{B}, B^{r}_{B}, and B^{a}_{B} are strength coefficients to be estimated.

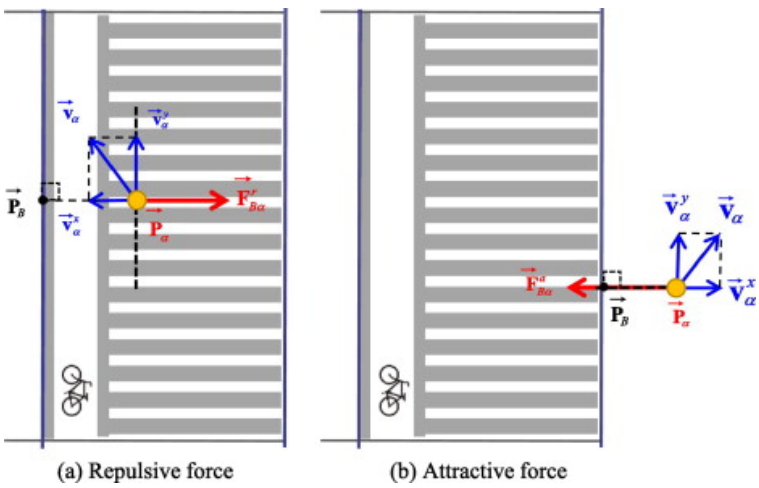

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Fig. 3. Repulsive and attractive forces from crosswalk boundary.

5.3. Force from surrounding pedestrians

As shown in Fig. 4, it is assumed that two kinds of collision avoidance behavior with surrounding pedestrians are related, i.e., group evasive and individual evasive behavior. When conflicting with others, pedestrians usually adjust their speed vectors to avoid conflict with the pedestrian group from the opposite direction first. This behavior may result in self-organization such as lane formation phenomena. Then, pedestrians will continue to avoid collision with the nearest conflicting pedestrian in the private sphere, which cause individual avoidance behavior such as deceleration or turning direction to keep a safe distance with each other. Thus, the force from surrounding pedestrians $\left(\vec{\mathbf{F}}_{P}\right)$ can be presented by Eq. (9).

$$\vec{\mathbf{F}}_{P} = \vec{\mathbf{F}}^{r}_{\alpha\beta} + \vec{\mathbf{f}}^{r}_{\alpha\beta} \tag{9}$$

where $\vec{\mathbf{F}}^{r}_{\alpha\beta}$ is the resultant force from the pedestrian group in the visual range, and $\vec{\mathbf{f}}^{r}_{\alpha\beta}$ is the force from the nearest pedestrian in the private sphere.

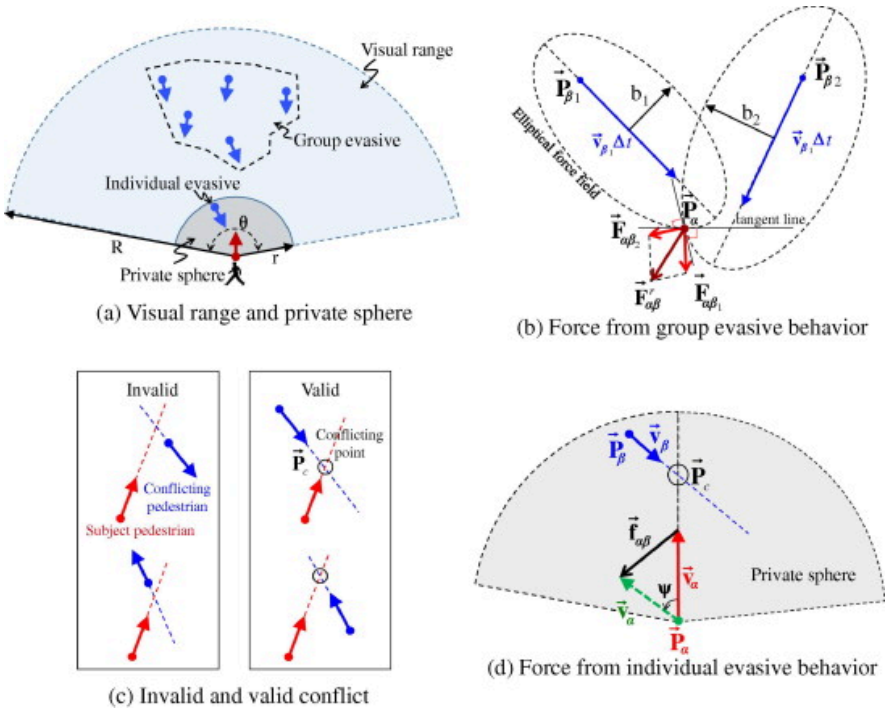

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Fig. 4. Force from surrounding pedestrians.

5.3.1. Group evasive behavior

As shown in Fig. 4(b), it is assumed that each conflicting pedestrian within the visual range will generate an elliptical force field resulting in the repulsive effect to the subject pedestrian. Referring to the classical social force model (Helbing and Molnar, 1995), the repulsive potential is an exponential decreasing function of b_i with equipotential lines having the form of an ellipse directed into the direction of motion. b_i is the length of the semi-minor axis of the elliptical force field generated by the conflicting pedestrian i. Thus, the resultant force from the conflicting pedestrian group can be presented by Eq. (10).

$$\vec{\mathbf{F}}^{r}_{\alpha\beta} = \sum_{i=1}^{n} A_\beta \exp(-B_\beta b_i)\vec{\mathbf{n}}_{\beta_i} \tag{10}$$

$$b_i = \frac{1}{2}\sqrt{\left(\left\|\vec{\mathbf{P}}_\alpha - \vec{\mathbf{P}}_{\beta_i}\right\| + \left\|\vec{\mathbf{P}}_\alpha - \vec{\mathbf{P}}_{\beta_i} - \vec{\mathbf{v}}_{\beta_i}\Delta t\right\|\right)^2 - \left(\left\|\vec{\mathbf{v}}_{\beta_i}\Delta t\right\|\right)^2}$$

where $\vec{\mathbf{p}}_\alpha$ is the current position of pedestrian α, $\vec{\mathbf{p}}_{\beta_i}$ is the current position of pedestrian β_i, $\vec{\mathbf{n}}_{\beta_i}$ is the normalized vector which is perpendicular to the tangent line of the elliptical force field of pedestrian β_i and the tangent point is $\vec{\mathbf{p}}_\alpha$. n is the number of conflicting pedestrians. A_β and B_β is the strength coefficient to be estimated.

5.3.2. Individual evasive behavior

It is assumed that the subject pedestrian will implement a giving-way maneuver (Asano et al., 2010) to the nearest conflicting pedestrian if someone enters the private sphere and a potential collision is going to occur. Fig. 4(c) shows two kinds of conflicts, however, only the valid conflict may lead to collision. The valid conflict can be identified by the time to conflict point (TTCP). TTCP is defined as the expected time to pass the conflict point if the two pedestrians remain at their current speeds and directions. The TTCP for the subject pedestrian α and the conflicting pedestrian β is given by Eq. (11). Such conflicts that the TTCPs of both pedestrians are positive will be deemed as valid, whereas a negative value means that one pedestrian have passed

the conflict point and no potential collision will occur.

$$TTCP_\alpha = \frac{\left\|\vec{\mathbf{P}}_c - \vec{\mathbf{P}}_\alpha\right\|}{\left(\left\|\vec{\mathbf{v}}_\alpha\right\|\right)} \cos\left\langle \vec{\mathbf{v}}_\alpha, \vec{\mathbf{P}}_c - \vec{\mathbf{P}}_\alpha \right\rangle$$
$$TTCP_\beta = \frac{\left\|\vec{\mathbf{P}}_c - \vec{\mathbf{P}}_\beta\right\|}{\left(\left\|\vec{\mathbf{v}}_\beta\right\|\right)} \cos\left\langle \vec{\mathbf{v}}_\beta, \vec{\mathbf{P}}_c - \vec{\mathbf{P}}_\beta \right\rangle \tag{11}$$

where $\vec{\mathbf{P}}_\beta$ is the current position of the conflicting pedestrian β, $\vec{\mathbf{P}}_c$ is the conflicting point position. The relative time to the conflicting point ($T_{\alpha\beta}$) in Eq. (12) can be used to evaluate the collision risk between two pedestrians. Larger values of $T_{\alpha\beta}$ mean lower collision risk.

$$T_{\alpha\beta} = \begin{cases} |TTCP_\alpha - TTCP_\beta|, & \text{if } TTCP_\alpha > 0 \quad \text{and} \quad TTCP_\beta > 0 \\ +\infty, & \text{otherwise} \end{cases} \tag{12}$$

Once a valid conflict is identified, a giving-way maneuver will be performed. As shown in Fig. 4(d), the subject pedestrian takes actions to avoid collisions with the nearest conflicting pedestrian β by adjusting the speed or direction. A lack of giving-way maneuver may lead to a problem that two pedestrians may stop forever (when $TTCP_\alpha \to +\infty$ and $TTCP_\beta \to +\infty$) due to social force balance. For instance, when two conflicting pedestrians in opposite directions move in a straight line, the social force balance may occur if no other disturbances come. Thus, an active mechanism to avoid such balance conditions is necessary. The giving-way maneuver is flexible and stochastic in the real world. Pedestrians can pass the nearest conflicting pedestrian smoothly by adopting either accelerating or decelerating maneuver. Meanwhile, direction changing or not is dependent on personal preference. To reflect the characteristic of the giving-way maneuver, a stochastic adjustment mechanism is developed. Fig. 5 shows the process of finding out a reasonable speed changing or direction changing to avoid the collision. If $T_{\alpha\beta}$ approaches infinity or is lower than an expected safety threshold of TTC (time to collision, assumed to be 2 s), a loop will start up to search a solution by adjusting the speed and direction randomly until $T_{\alpha\beta}$ reaches to the safety threshold of TTC. The adjusted speed vector $\left(\vec{\mathbf{v}}^*_\alpha\right)$ can be presented by Eq. (13).

$$\vec{\mathbf{v}}^*_\alpha = (1 + \Delta\nu)\vec{\mathbf{v}}_\alpha \boldsymbol{\omega}(\psi)$$
$$\boldsymbol{\omega}(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix} \tag{13}$$

where $\boldsymbol{\omega}(\psi)$ is the anticlockwise rotation operator. Accordingly, the individual repulsive force $\left(\vec{\mathbf{f}}^r_{\alpha\beta}\right)$ from the nearest conflicting pedestrian can be presented by Eq. (14).

$$\vec{\mathbf{f}}^r_{\alpha\beta} = \frac{1}{\Delta t}\left(\vec{\mathbf{v}}^*_\alpha - \vec{\mathbf{v}}_\alpha\right) \tag{14}$$

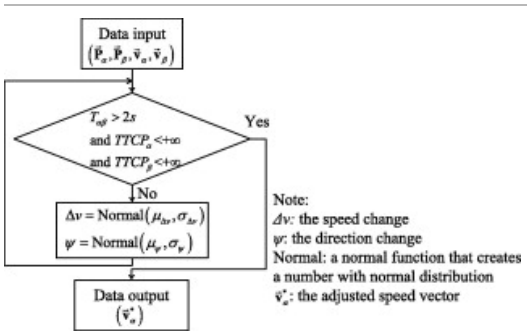

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Fig. 5. Flow chart of individual collision avoidance.

5.4. Force from left-turning vehicle

Since it is left-hand traffic in Japan, left-turning vehicles usually conflict with pedestrians at a permitted phase. Generally, drivers are required to yield to crossing pedestrians. However, some drivers may take risky behavior by accepting short gaps and threaten the safety of pedestrians. It is assumed that pedestrians are affected by the vehicle force field once he/she approaches it. As shown in Fig. 6, the subject pedestrian will take actions to avoid collision with the left-turning vehicle although the vehicle may yield to the pedestrian. This situation is similar to collision avoidance with conflicting pedestrians, though the physical size of a left-turning vehicle is much larger than that of a pedestrian. Similarly, an elliptical force field is assumed to describe the force generated by the nearest left-turning vehicle. The difference from the force field of a conflicting pedestrian is that the two focuses of the ellipse are defined by the vehicle near-side rear position $\left(\vec{\mathbf{P}}^{*\,near}_{rear}\right)$ and the predicted vehicle near-side head position $\left(\vec{\mathbf{P}}^{*\,near}_{head}\right)$ in next time step. Thus, the repulsive force from a left-turning vehicle can be presented by Eq. (15).

$$\vec{\mathbf{F}}_V = A_\nu \exp(-B_\nu b_\nu)\vec{\mathbf{n}}_{\nu\alpha} \tag{15}$$

where A_v and B_v are the strength coefficients, $\vec{\mathbf{n}}_{\nu\alpha}$ is the normalized vector perpendicular to the tangent line of the elliptical force field of the left-turning vehicle, b_v is the length of semi-minor axis of the elliptical force field, which can be presented by Eq. (16).

$$b_\nu = \frac{1}{2}\sqrt{\left(\left\|\vec{\mathbf{P}}_\alpha - \vec{\mathbf{P}}^{near}_{rear}\right\| + \left\|\vec{\mathbf{P}}_\alpha - \vec{\mathbf{P}}^{*\,near}_{head}\right\|\right)^2 - \left(\left\|\vec{\mathbf{P}}^{near}_{rear} - \vec{\mathbf{P}}^{*\,near}_{head}\right\|\right)^2} \tag{16}$$

where $\vec{\mathbf{P}}^{*\,near}_{head}$ can be presented by Eq. (17).

$$\vec{\mathbf{P}}^{*\,near}_{head} = \vec{\mathbf{P}}^{near}_{head} + \vec{\mathbf{v}}_{veh}\Delta t \tag{17}$$

where $\vec{\mathbf{v}}_{veh}$ is the current speed vector of the conflicting vehicle.

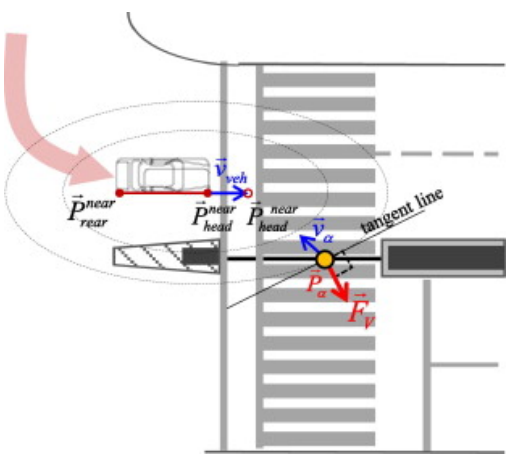

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Fig. 6. Force from the conflicting left-turning vehicle.

5.5. Attractive force from signal

Pedestrian behavior is also influenced by the attractive force from signal control. Empirical analyses indicate that pedestrian speeds are significantly higher at pedestrian flashing green (PFG) phase than those at pedestrian green phase (PG). The major reason is that pedestrians need to finish crossing before the onset of the red signal as soon as possible. Here, it is assumed that if the pedestrian is still at the crosswalk when the signal has changed to PFG or Red, he/she will accelerate and cross as fast as possible. As shown in Fig. 7, the magnitude of the force is assumed to linearly relate to the current distance to the desired exit position. The farther pedestrians are from the desired exit position, the stronger the force will be. However, if the current speed has achieved the maximum acceptable speed, the force should be zero. Thus, the attractive force $\left(\vec{\mathbf{F}}_S\right)$ from the signal can be

presented by Eq. (18).

$$\vec{\mathbf{F}}_S = \begin{cases} \left(A_s\left\|\vec{\mathbf{P}}_\alpha - \vec{\mathbf{P}}_{exit}\right\| + B_s\right)\vec{\mathbf{e}}_\alpha, & \text{if the signal phase is PFG or Red, and } \left\|\vec{\mathbf{v}}_\alpha\right\| < v_{\max} \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

where A_s and B_s are the strength coefficients, $\vec{\mathbf{P}}_{exit}$ is the desired exit position of the subject pedestrian α, $v_{\max}$ is the pedestrian maximal acceptable speed.

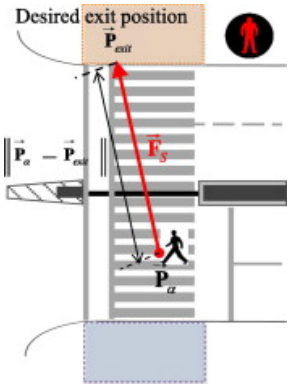

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Fig. 7. Attractive force from signal.

6. Model calibration

6.1. Data acquisition

As shown in Fig. 8, the north crosswalk at Kanayama intersection in Nagoya City of Japan was chosen as study site. The positions of pedestrians and left-turning vehicles were manually extracted from the video every 0.5 s by using an image processing software (Suzuki and Nakamura, 2006). The dataset consists of the trajectories of 1352 pedestrians and 124 left-turning vehicles. In total, 97,200 position samples are available for trajectory profiles based on time series. From these data, all relevant variables can be derived either directly or by applying finite differences, such as positions $\left(\vec{\mathbf{P}}_\alpha(t_k)\right)$, velocities $\left(\vec{\mathbf{v}}_\alpha(t_k)\right)$, accelerations $\left(\vec{\mathbf{a}}_\alpha(t_k)\right)$, distances between pedestrians, and speed angles between pedestrians.

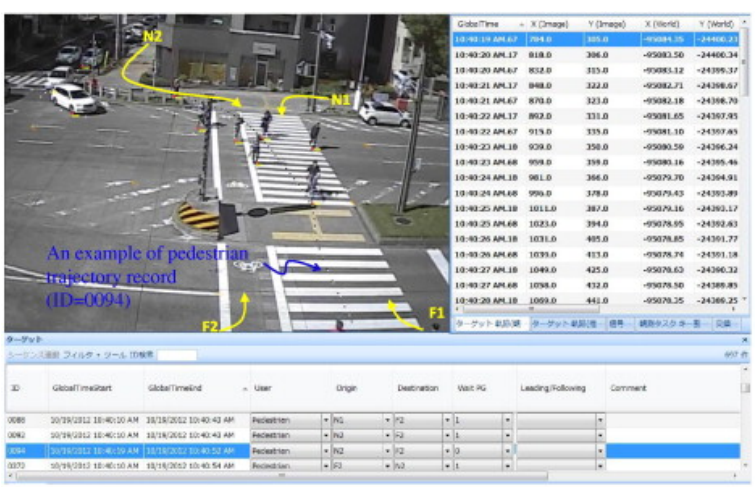

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Fig. 8. Test bed for data acquisition.

6.2. Calibration methodology

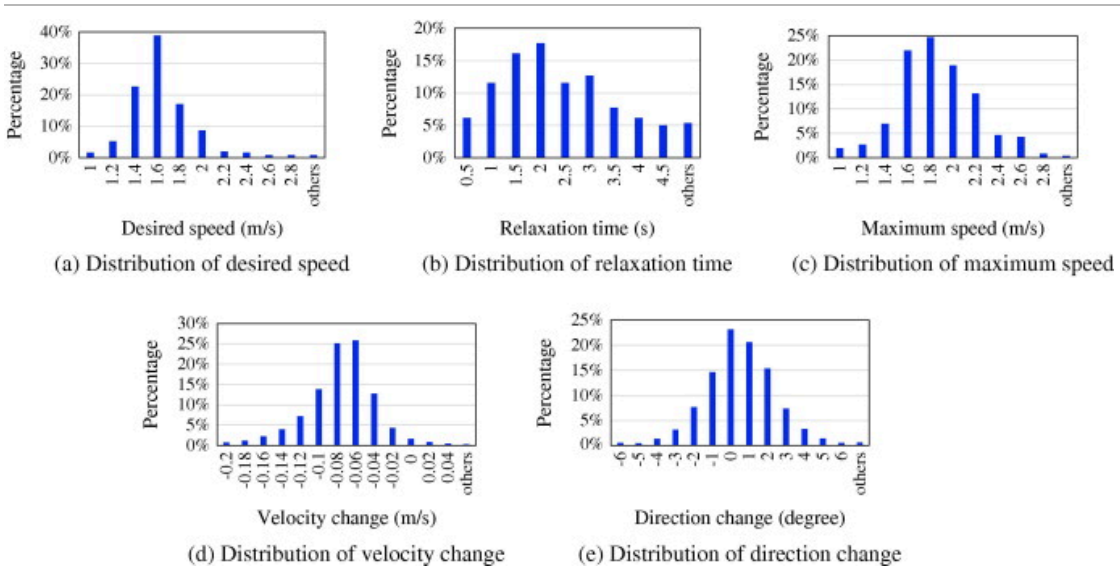
At the early stage, model parameters are commonly calibrated by comparing aggregate outcomes such as speed–density relationship (Seyfried et al., 2006), or emerging pedestrian movement patterns such as lane formation (Helbing et al., 2000). However, it is still unclear whether a microscopic model is able to represent individual walking behavior accurately with a macroscopic calibration method, considering the fact that aggregate outcomes only provide a reasonable average prediction (Hoogendoorn and Daamen, 2006). Recently, as the calibration of microscopic pedestrian models has been considered as an important issue, some researchers have calibrated social force models by the maximum log-likelihood estimation method which is able to describe the parameters in a statistical way (Daamen and Hoogendoorn, 2012, Ko et al., 2013). The calibration methodology adopted in this study is similar to the one used by Hoogendoorn and Daamen (2006). However, there exist two major differences. First, considering that acceleration is the second difference of the position, to estimate acceleration will amplify the error in trajectory processing. Thus, this study uses the raw positions of trajectory rather than the estimated acceleration for calibration. Secondly, the parameters are separated into two groups, i.e., measurable parameters and non-measurable parameters, and calibrated respectively.

The developed social force model contains various parameters. As some of them cannot be derived from the observed dataset, direct calibration is difficult. Here, we adopt a three-stage process to alleviate this problem. As the first step, the parameters such as visual range, which are measurable but difficult to be identified from the trajectory dataset, are estimated by referring to related studies. Secondly, the parameters such as relaxation time, which are possible to be identified from the observed dataset, are directly estimated. Finally, after fixing the above parameters, other parameters, such as the strength coefficients of each force that do not have concrete physical meanings, are derived by MLE.

6.2.1. Calibration of measurable parameter

Altogether, there are seven measurable parameters in the developed social force model, i.e., angle of visual range (θ), radius of visual range (R), desired speed ($\boldsymbol{v}_{\alpha}^{d}$), relaxation time ($\tau_{\alpha}$), maximum acceptable speed ($v_{\max}$), velocity change (Δv) and direction change (ψ). The visual range is defined as the radius and angle at which a given standard object can be seen with unaided eyes. Due to the limitation of data collection, it is difficult to directly estimate the visual range based on the trajectory dataset. Here, according to related studies (Guo et al., 2010, Antonini et al., 2006), the angle of visual range is set as 170° and the radius of visual range is set as 8 m.

Desired speed is defined as the personal average speed if a pedestrian crosses the road without any disturbance by signal, boundary, surrounding pedestrians and vehicles. The trajectory data were collected when there were no conflicting pedestrians and left-turning vehicles at pedestrian green phases and the subject pedestrian must be inside the crosswalk, which guarantees less repulsive or attractive force influencing the subject pedestrian. As shown in Fig. 9(a), the results show that the average value of desired speed is 1.6 m/s and the standard deviation is 0.15 m/s.



(a) Distribution of desired speed (b) Distribution of relaxation time (c) Distribution of maximum speed

(d) Distribution of velocity change (e) Distribution of direction change

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Fig. 9. Calibration results of measurable parameters.

Relaxation time is defined as the time for a pedestrian recovering from current speed to the personal desired speed without any disturbance. To reduce the influence of other forces, only the dataset for desired speed estimation was used. However, the pedestrian speeds still change dynamically due to personal random decision. Here, the relaxation time is estimated as the acceleration period from a lower current speed to the personal desired speed. As shown in Fig. 10, if the current speed is lower than the desired speed, a loop will start up to search forward and find out how many time steps were needed until the speed recovers to the desired speed. As shown in Fig. 9(b), the average value is 2.2 s and the standard deviation is 0.5 s.

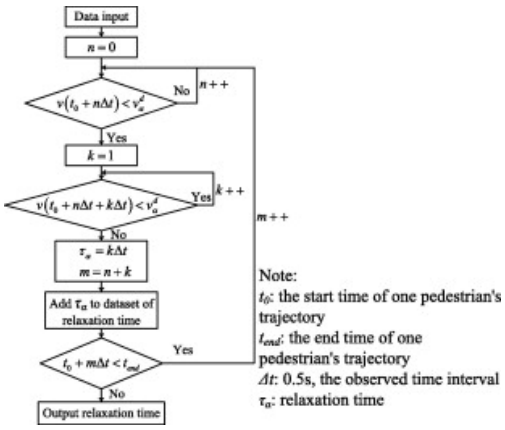

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Fig. 10. Flow chart of relaxation time estimation.

Maximal acceptable speed is defined as the maximum instant speed along one pedestrian trajectory. Here, all of the trajectories are utilized as the estimation dataset. As shown in Fig. 9(c), the average of maximal acceptable speeds is 1.8 m/s and the standard deviation of maximal acceptable speeds is 0.3 m/s.

Velocity change is defined as the speed difference between the quantities of the current speed and the speed at next time interval when the subject pedestrian interacts with the nearest conflicting pedestrian in the private sphere. To reduce other influences, the trajectory data collection meets the following requirements. First, there was only one conflicting pedestrian and he/she was in the private sphere of the subject pedestrian. Second, no left-turning vehicles exist. Third, it was at pedestrian green phases. Finally, the subject pedestrian must be inside the crosswalk. As shown in Fig. 9(d), the average value is −0.07 m/s and the standard deviation is 0.09 m/s.

Direction change is the angle difference between the current speed vector and the speed at next time interval when the subject pedestrian interacts with the nearest conflicting pedestrian in the private sphere. The data collection condition is the same as the condition of velocity change. As shown in Fig. 9(e), the average value is 0.03° and the standard deviation is 2.14°.

6.2.2. Calibration of non-measurable parameters

After fixing the estimated measurable parameters, other parameters which are difficult to measure directly, i.e., A_β, B_β, A_B^r, A_B^a, B_B^r, B_B^a, A_v, B_v, A_s and B_s, are calibrated by MLE. The position for the next time step $\left(\vec{\mathbf{P}}_\alpha(t_{k+1}|\boldsymbol{\theta}_p)\right)$ closely depends on the model parameters ($\boldsymbol{\theta}_p$) to be estimated. The estimated distance vector $\left(\Delta\vec{\mathbf{d}}_\alpha^{est}(t_k|\boldsymbol{\theta}_p)\right)$ is pointing from $\vec{\mathbf{P}}_\alpha(t_k|\boldsymbol{\theta}_p)$ to $\vec{\mathbf{P}}_\alpha(t_{k+1}|\boldsymbol{\theta}_p)$. Here, it is assumed that the length of $\Delta\vec{\mathbf{d}}_\alpha^{est}(t_k|\boldsymbol{\theta}_p)$ in x direction and in y direction are normally distributed with mean (μ_x,μ_y) and standard deviation (σ_x, σ_y). (μ_x, μ_y) and (σ_x, σ_y) can be estimated from the observed distance vector $\left(\Delta\vec{\mathbf{d}}_\alpha^{obs}(t_k)\right)$. The likelihood L_k of a single prediction step, i.e., from time t_k to t_{k+1}, is directly related to the probability density function of the normal distribution as shown in Eq. (19).

$$L_k(\Delta d_\alpha(\boldsymbol{\theta}_p)|\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-(\Delta d_\alpha(\boldsymbol{\theta}_p)-\mu)^2}{2\sigma^2}} \tag{19}$$

Note that the distance vector $\left(\Delta\vec{\mathbf{d}}_\alpha^{obs}(t_k)\right)$ is a two-dimensional variable. To simplify the calculation, it is assumed that both vector components in x direction and in y direction are normal distributions. Thus, $\Delta d_\alpha(\boldsymbol{\theta}_p)$ is the combination of $\Delta\vec{\mathbf{d}}_{\alpha x}^{est}(t_k|\boldsymbol{\theta}_p)$ and $\Delta\vec{\mathbf{d}}_{\alpha y}^{est}(t_k|\boldsymbol{\theta}_p)$ as shown in Eq. (20). Accordingly, μ and σ in Eqs. (21), (22) are derived from (μ_x, μ_y) and (σ_x, σ_y) respectively.

$$\Delta d_\alpha(\boldsymbol{\theta}_p) = \left\|\Delta\vec{\mathbf{d}}_{\alpha x}^{est}(t_k|\boldsymbol{\theta}_p)\right\| + \left\|\Delta\vec{\mathbf{d}}_{\alpha y}^{est}(t_k|\boldsymbol{\theta}_p)\right\| \tag{20}$$

$$\mu = \mu_x + \mu_y \tag{21}$$

$$\sigma = \sqrt{\sigma_x^2 + \sigma_y^2 + 2\mathrm{cov}\left(\left\|\Delta\vec{\mathbf{d}}_{\alpha x}^{est}(t_k|\boldsymbol{\theta}_p)\right\|, \left\|\Delta\vec{\mathbf{d}}_{\alpha y}^{est}(t_k|\boldsymbol{\theta}_p)\right\|\right)} \tag{22}$$

where $\Delta\vec{\mathbf{d}}_{\alpha x}^{est}(t_k|\boldsymbol{\theta}_p)$ is the vector components $\Delta\vec{\mathbf{d}}_\alpha^{est}(t_k|\boldsymbol{\theta}_p)$ in x direction, and $\Delta\vec{\mathbf{d}}_{\alpha y}^{est}(t_k|\boldsymbol{\theta}_p)$ is the vector components $\Delta\vec{\mathbf{d}}_\alpha^{est}(t_k|\boldsymbol{\theta}_p)$ in y direction. Considering the entire observed samples and neglecting the correlation between subsequent samples, the likelihood of the observation given for the model parameters ($\boldsymbol{\theta}_p$) can be presented by Eq. (23).

$$L(\Delta d_\alpha(\boldsymbol{\theta}_p)|\mu,\sigma) = \prod \frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-(\Delta d_\alpha(\boldsymbol{\theta}_p)-\mu)^2}{2\sigma^2}} = \frac{(2\pi)^{-\frac{n}{2}}}{\sigma^n}\exp\left[-\frac{\sum(\Delta d_\alpha(\boldsymbol{\theta}_p)-\mu)^2}{2\sigma^2}\right] \tag{23}$$

To facilitate the computation, the logarithm is taken on both sides of Eq. (19). Thus, the log-likelihood can be presented by Eq. (24). The maximum log-likelihood estimates of model parameters ($\boldsymbol{\theta}_p$) are obtained such that Eq. (24) is maximized.

$$\ln L = -\frac{1}{2}n\ln(2\pi) - n\ln(\sigma) - \frac{\sum(\Delta d_\alpha(\boldsymbol{\theta}_p)-\mu)^2}{2\sigma^2} \tag{24}$$

Table 1 shows the calibration results of parameter estimation. According to p values at the 95% confidence level, all the parameters in the modified social force model are significant. The estimation results show that the coefficient for the interaction strength of vehicle repulsive force has the largest value, implying a higher sensitivity of collision avoidance with vehicles on pedestrian crossing behavior.

Table 1. Calibration results of non-measurable parameters.

Parameters	Eq.	Estimates	*p* Value	Description
A_B^r	(8)	0.21	0.00	Strength coefficients for repulsive force from crosswalk boundary
B_B^r	(8)	0.84	0.00	
A_B^a	(8)	0.42	0.01	Strength coefficients for attractive force from crosswalk boundary

Parameters	Eq.	Estimates	*p* Value	Description
B_B^a	(8)	0.95	0.00	
A_β	(10)	0.85	0.00	Strength coefficients for repulsive force from surrounding conflicting pedestrians
B_β	(10)	0.76	0.01	
A_v	(15)	1.32	0.02	Strength coefficients for repulsive force from conflicting left-turning vehicle
B_v	(15)	0.86	0.00	
A_s	(18)	0.05	0.02	Strength coefficients for attractive force from PFG and Red phase signal
B_s	(18)	0.08	0.00	

7. Model performance

To verify the performance, the developed model was incorporated into a microscopic simulation platform (Dang et al., 2012), in which the vehicle behavior model had been incorporated. The pedestrian behavior is simulated for a large number of interacting pedestrians confronted with different situations at signalized intersection.

7.1. Individual trajectory in simulation

To analyze the performance of the developed model, one of the pedestrian trajectories was extracted from the observed and estimated dataset, respectively. The sample pedestrian was simulated while keeping the other pedestrians on their original trajectories.

As shown in Fig. 11(a), the force from surrounding pedestrians has the most significant influence (an average of about 66.1% of the resultant force during the whole period) on the pedestrian behavior because the tested pedestrian conflicts with others during the whole trip. The force from vehicle is zero at the beginning. However, from the 8th second to the 14th second, the subject pedestrian conflicts with a left-turning vehicle. During this period, the force from conflicting vehicle accounts for an average of 15.3%. The driving force is around 18.5% of the resultant force during the whole period, providing the power to recover to the desired speed toward destination. The force from boundary is about 12.4% of the resultant force, which is relatively stable during the whole crossing process. The force from signal is zero because the crossing behavior happened at pedestrian green phase.

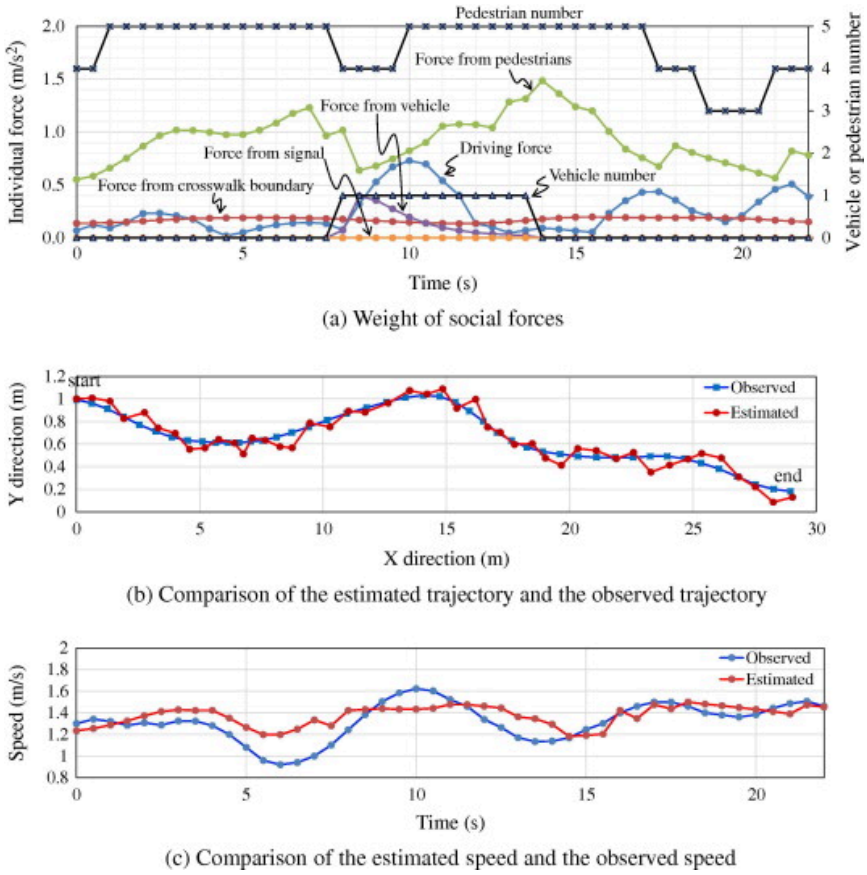

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Fig. 11. Verification of estimated pedestrian trajectory.

Fig. 11(b) shows the comparison between the estimated and observed positions of the total crossing trajectory. It was found that the *Mean Absolute Percentage Error* (*MAPE*) of the position estimation in each time step is 5.7%. Fig. 11(c) shows the comparison between the estimated and observed speeds in time series. The *MAPE* of speed estimation is 10.5%, a little bit larger than that of position estimation. One possible reason is that speed is the difference of the position and further amplifies the estimation error.

7.2. Passing position at three cross-sections

Fig. 12 shows the distribution of observed and simulated pedestrian trajectories from N2 to F2 at Kanayama North crosswalk. Note that N1, N2, F1 and F2 denote the OD of pedestrian movement. No significant difference of passing positions is found at the three representative cross-sections according to two sample *t*-test for difference of their means at the 95% confidence interval. In addition, it indicates that the distribution of passing position is closely related to pedestrian OD movements. For example, in the case of near-side passing position, pedestrians with the origin N2 tend to enter the crosswalk close to the left boundary of crosswalk (near N2). Then, the peak of the histogram at middle cross-section shifts a little bit to the right side, indicating that pedestrians have change their speed vectors when interacting with other road users. Finally, the peak of the histogram at far-side cross-section further shifts to the left since pedestrians are appealed by the destination (F2). This validation demonstrates that the desired direction model and driving force model can reasonably reflect the pedestrian behavior characteristic at crosswalk.

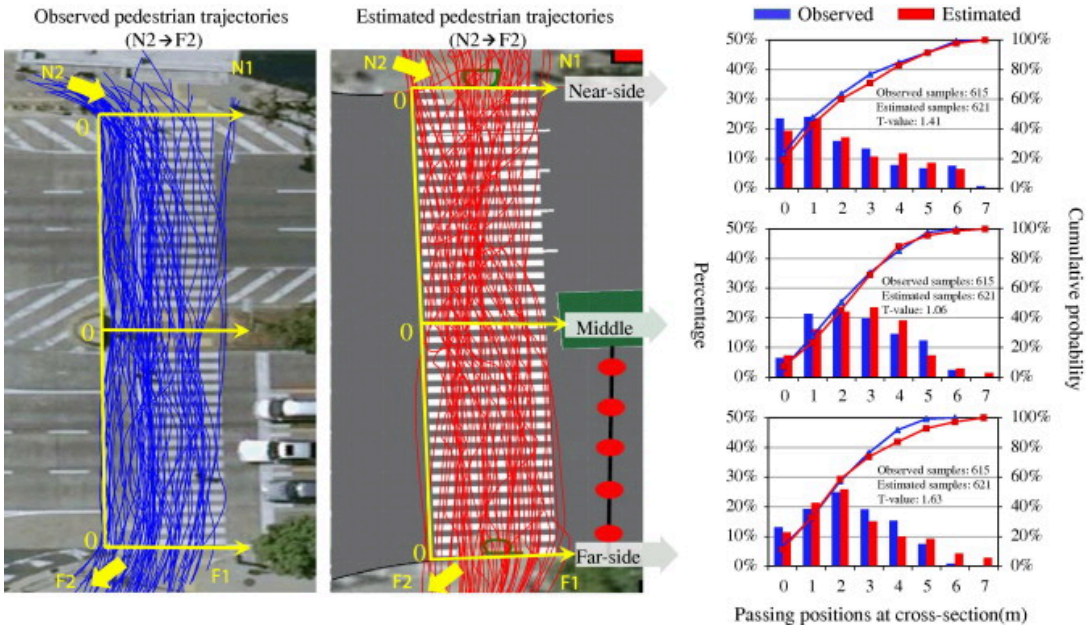

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Fig. 12. Verification of estimated pedestrian passing positions at three cross-sections.

7.3. Collision avoidance with turning vehicle

To analyze the interaction with left-turning vehicles, the pedestrian passing position at a specified cross-section is chosen for validation under two following conditions. One is pedestrian vs. vehicle conflicting condition with turning vehicles existing at or approaching the crosswalk when pedestrians are crossing, whereas the other is no conflict condition. The specified cross-section, i.e., the central line of the third vehicle lane (from the curb of the exit approach), is chosen for analysis, considering the pedestrian vs. vehicle conflict occurs most frequently along this exit lane according to the observed dataset. Fig. 13 shows the comparison of observed and estimated passing positions at the specified cross-section. It is shown that the peak of the histogram in observed dataset shifts from 0 to 2 m if the pedestrian comes across left-turning vehicles. This phenomenon indicates that pedestrians are likely to detour to avoid collision with vehicles. According to *t*-test at the 95% confidence interval, no significant difference was found between observed and estimated passing positions. It demonstrates that the developed model can well present the behavior of collision avoidance with vehicles. Note that the peak of the histogram in estimated dataset shifts from 0 to 2.5 m, indicating that the reaction of pedestrian in simulation is more sensitive than that in real situation. Thus, the repulsive force from turning vehicle still needs to be improved in the future development of the model.

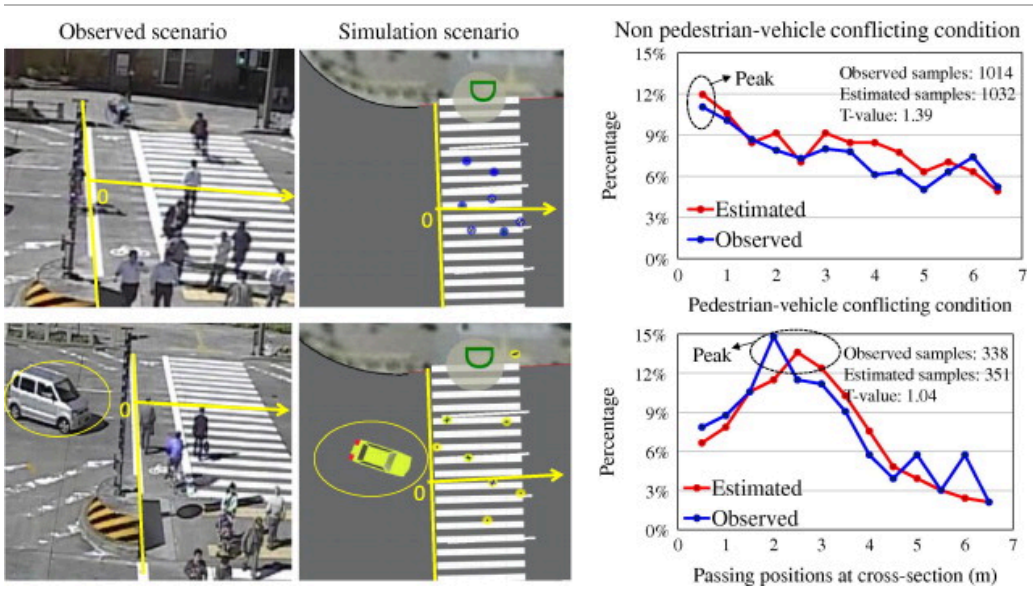

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Fig. 13. Verification of collision avoidance with turning vehicles.

7.4. Self-organization phenomenon

Self-organization is one of the important indicators to verify whether the simulation actually produces real world phenomena or not. A representative self-organization phenomenon is lane-formation. Due to the impact of various forces from surrounding environment, the lanes of pedestrian flow will form dynamically and their numbers may depend on OD movement, pedestrian density and crosswalk width, etc. Two simulation models were applied to examine the lane-formation phenomenon. As shown in Fig. 14(a), one is route-choice based desired direction model, given that the individual decision-making of desired direction is like the route-choice strategy. The desired direction is determined by Eq. (1) which is related to various influencing factors such as OD movement and pedestrian density. The other is non-route-choice based desired direction model. As shown in Fig. 14(b), without considering any influence from surrounding environment, a uniform random distribution is applied to generate the desired direction.

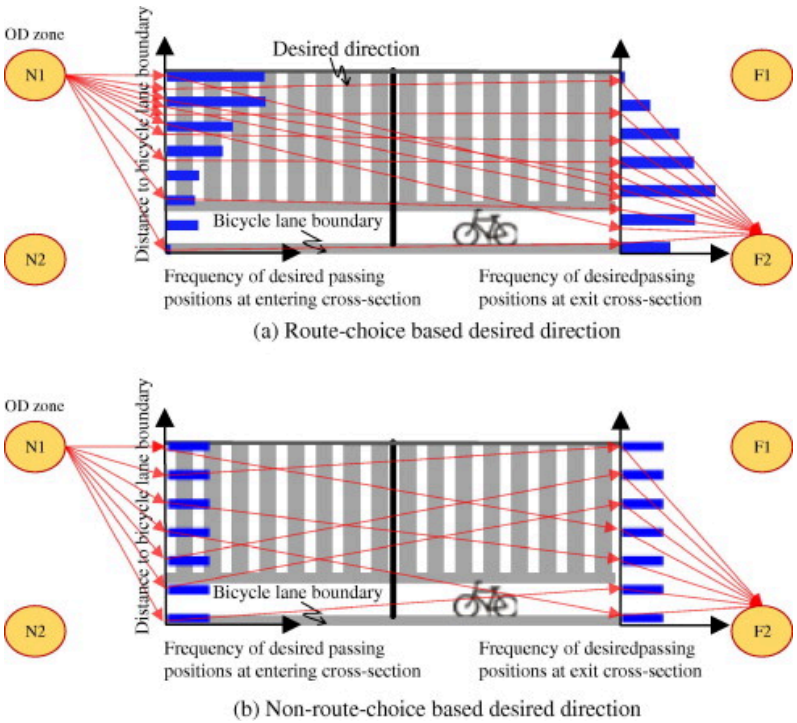

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Fig. 14. Desired direction settings (example: N1→F2).

Fig. 15 shows the lane-formation phenomenon in the real world and in simulation. It demonstrates that the modified social force model is capable of reflecting the significant self-organization phenomenon in simulation, though the shape of the formed lanes is not exactly the same. Comparing to the lane-formation by using non-route-choice based model, the pedestrian group in the same moving direction develops with a larger size and the length of the formed pedestrian lane appears longer by using route-choice based model. Moreover, the lane forms easier and keeps a relatively stable shape for a longer period when the route-choice based model is applied. In contrast, the formed lanes by using non-route-choice based model are smaller and unstable, which are easier to break down.

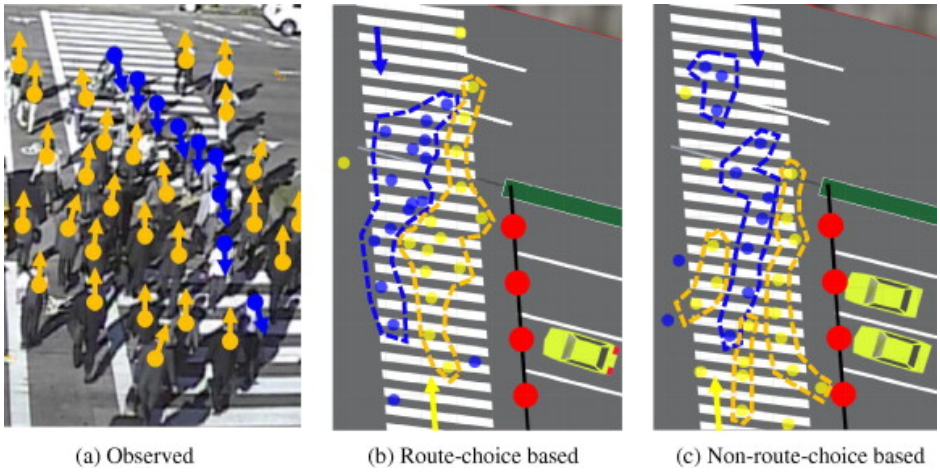

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Fig. 15. Lane-formation phenomenon in the real world and in simulation.

To compare the pedestrian flow performance between the route-choice based model and non-route-choice based model, various performance indices such as the number of formed lanes, the instant angle between current direction and desired direction, and the passing time at crosswalk, are analyzed.

The comparison of the number of formed lanes is shown in Fig. 16(a). The number of formed lanes was collected manually when two opposite pedestrian flows begin to come across and the segregation phenomenon occurs. The average number of formed lanes is 2.8 in observed dataset while it is 3.5 in simulation by using route-choice based model and 4.1 by using non-route-choice based model. It demonstrates that the number of formed lanes by route-choice based model is closer to the observed value.

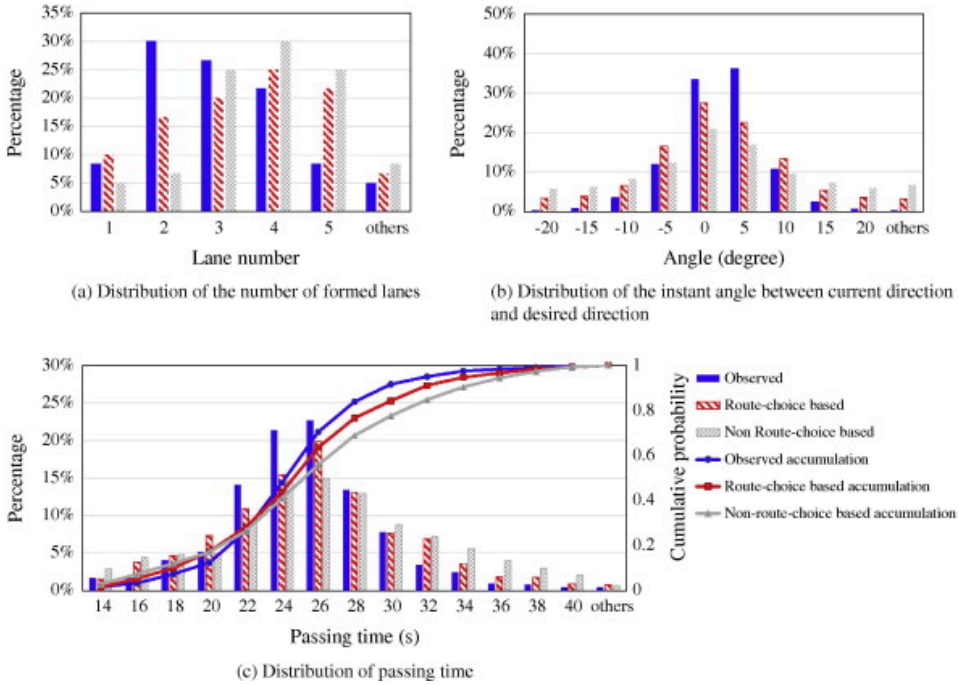

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Fig. 16. Pedestrian flow performance.

Fig. 16(b) shows the distribution of the instant angle between current direction and desired direction. The instant angle was collected in each time step. Zero degree means the pedestrian move toward the desired passing position straight. The angle variation can reflect the frequency of interactions with conflicting pedestrians or other factors. According to the simulation result, most of the pedestrians keep their directions toward the desired destination or only change a small angle in each time step, which agrees with the statistics in observed dataset. However, the angle variation is larger by using non-route-choice model. It demonstrates that the interaction among conflicting pedestrians is relatively severe because of the unstable lane-

formation. The performance of instant angle distribution is closer to the real situation by using route-choice based model, since the desired direction can be reproduced by a reasonable distribution function in simulation, which helps to represent lane-formation phenomenon as in the real world.

Fig. 16(c) shows the distribution of the passing time at crosswalk. It was found that there are about 90% of the pedestrians finishing crossing within 30s in observed dataset while it is about 86% by using route-choice based model and 76% by using non-route-choice based model in simulation. The performance of passing time gets improved by using route-choice based model. One possible reason can be attributed to the fact that the lane-formation is more stable. In reality, it is easier for pedestrians to cross the road by following someone in lane-formation situation than moving against opposite flows. Therefore, the passing times by using route-choice based model are shorter when compared to those by using non-route-choice based model.

8. Conclusions and future work

Based on pedestrian psychological process and the social force theory, a modified social force model considering the typical pedestrian behavior at signalized intersection, e.g., group evasive and individual evasive maneuver, collision avoidance with turning vehicles, and reaction to crosswalk boundary was developed. Then, the parameters of the developed model were calibrated in a statistical way. The measurable parameters were directly estimated based on the observed data and the non-measurable ones were derived by maximum likelihood estimation. Finally, the model performance was verified by various terms, such as instant position, instant speed, pedestrian trajectory, collision avoidance behavior with conflicting vehicles, and lane-formation phenomenon. The major contributions of this study are summarized as follows.

(1) The desired direction model was developed for driving force. Most of the studies so far commonly assumed simplified directions of OD as desired directions and the influencing factors such as crosswalk geometry and pedestrian density were seldom considered. This study developed a desired direction model by estimating the desired exit position at the edge of crosswalk, which enables to represent the pedestrian maneuver toward a predetermined position more realistically. It is considered as an important step to approximate the pedestrian trajectory at crosswalk. The verification of passing positions at three cross-sections showed no significant difference between estimated and observed trajectories, which demonstrated that the desired direction model and driving force model can reasonably reflect the pedestrian behavior characteristic at crosswalk.

(2) The typical stop/go decision-making behavior at PFG phase was incorporated, whereas the classical social force model paid less attention upon this. The model was intended to represent the risk-taking crossing behavior at PFG, which is crucial to pedestrian safety assessment at signalized intersections.

(3) Not only repulsive effect, but also attractive effect were considered in the force from crosswalk boundary. Considering that the crosswalk boundary is different from a rigid wall, pedestrians may walk outside the boundary when the conflicts with other pedestrians become severe due to increasing density. Classical social force from the border only presents the repulsive effect, which makes the pedestrians always inside a walkway. The newly developed force from crosswalk boundary not only maintains the repulsive effect, but also provides an attractive effect to those pedestrians outside the crosswalk.

(4) A giving-way maneuver was modeled for individual evasion. The stochastic adjustment mechanism by speed changing and direction changing was developed, which enables to effectively solve the problems in simulation that two conflicting pedestrians may collide or stop forever ($TTCP_{\alpha} \rightarrow +\infty$ and $TTCP_{\beta} \rightarrow +\infty$) due to social force balance.

(5) The force from conflicting left-turning vehicle was modeled. Avoiding collision with conflicting vehicle is one of the typical pedestrian behavior at crosswalk. Similar to the repulsive force from pedestrian group, an elliptical force field was assumed to describe the force generated by the nearest left-turning vehicle. The verification of collision avoidance with turning vehicle showed that the distribution of passing position at a specified cross-section is able to reflect the repulsive effect by conflicting vehicle.

(6) The attractive force from PFG or Red phase was developed. This force is generated when the signal changes to PFG or Red. The magnitude of the force was assumed to linearly relate to the current distance to the desired exit position. The farther pedestrians are from the desired exit position, the stronger the force will be. This attractive force can reflect the real phenomenon that pedestrians prefer to finish crossing as soon as possible when he/she perceives potential risk.

(7) In addition, this study provides a more detailed procedure for model calibration by distinguishing measurable and non-measurable parameters. It was demonstrated to be an effective way to achieve favorable calibration results.

(8) The analysis of lane-formation phenomenon demonstrates that the modified social force model is capable of reflecting the typical characteristic of self-organization in simulation. Moreover, the desired direction model proposed by this study helps to improve the pedestrian flow performance at crosswalk.

However, there are still several problems needing further attention in future work. For example, the reaction of pedestrians to vehicles is more sensitive in simulation than in real world. In addition, the pedestrian heterogeneous characteristics such as age, gender, and partner relationship should also be considered as the influencing factors of crossing behavior. These limitations will be improved by a more elaborate modeling approach and a more strict calibration procedure in our future work.

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